Comparison of Physics-Based, GRU and PhyGRU Models for Vehicle Lateral Acceleration Prediction

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Abstract

This work investigates the prediction of lateral acceleration in vehicle dynamics using three alternative recurrent modeling strategies: a purely data-driven GRU, a classical physics-based bicycle model, and a residual PhyGRU architecture. The objective is to assess how explicit physical structure and learnable temporal correction affect predictive performance on a nonlinear lateral vehicle simulation. The considered dataset is generated from a saturated bicycle-type vehicle model with steering and longitudinal speed as inputs, while the prediction target is lateral acceleration. The study compares model accuracy across multiple training trajectory lengths, highlighting the trade-off between interpretability, inductive bias, and data-driven flexibility in dynamic state estimation.

1 Introduction

Lateral acceleration is a central variable in vehicle dynamics and chassis control because it is directly related to handling, stability, and tire-road interaction. It is used in electronic stability control, vehicle state estimation, motion control, rollover prevention, and driver-assistance functions. In practical settings, direct measurement may be noisy, delayed, unavailable, or affected by sensor placement, which motivates accurate model-based or learning-based prediction.

Estimating lateral acceleration from steering input and longitudinal speed is therefore a relevant engineering task, especially when the goal is to capture transient behavior under varying operating conditions. From a modeling perspective, the problem is also interesting because lateral vehicle dynamics are governed by nonlinear effects such as tire saturation, coupling between lateral velocity and yaw rate, and dependence on longitudinal velocity. These features make the task suitable for comparing a classical physics model, a fully data-driven recurrent network, and a hybrid physics-informed recurrent architecture.

In this work, the same prediction problem is studied under different trajectory lengths in order to understand how much temporal training set size is needed for each modeling paradigm to generalize effectively. The study is motivated by data-scarce engineering settings, where partial physical knowledge is available and hybrid modeling can be especially valuable.

2 Problem setup

At each discrete time step t, the input is

$$u_t = \begin{bmatrix} \delta_t \\ v_{x,t} \end{bmatrix},$$

where δ_t is steering angle and $v_{x,t}$ is longitudinal speed. The prediction target is the lateral acceleration

$$y_t = a_{y,t}.$$

For a sequence of length T, the learning problem is

$$\{u_t\}_{t=1}^{T} \mapsto \{\hat{y}_t\}_{t=1}^{T}.$$

The synthetic vehicle used to generate the data follows a nonlinear lateral model with state

$$x_t = \begin{bmatrix} v_{y,t} \\ r_t \end{bmatrix},$$

where v_y is lateral velocity and r is yaw rate. With wheelbase parameters l_f, l_r, mass m, yaw inertia I_z, and cornering gains c_f, c_r, the slip angles are

$$\alpha_f = \delta_t - \frac{v_{y,t} + l_f r_t}{v_{x,t}}, \qquad \alpha_r = -\frac{v_{y,t} - l_r r_t}{v_{x,t}}.$$

The tire forces are saturated with a smooth nonlinearity,

$$F_{yf} = F_f^{\max} \tanh\left(\frac{c_f}{F_f^{\max}} \alpha_f\right), \qquad F_{yr} = F_r^{\max} \tanh\left(\frac{c_r}{F_r^{\max}} \alpha_r\right),$$

with

$$F_f^{\max} = \mu m g \frac{l_r}{l_f + l_r}, \qquad F_r^{\max} = \mu m g \frac{l_f}{l_f + l_r}.$$

The continuous-time dynamics are

$$\dot{v}_y = \frac{F_{yf} + F_{yr}}{m} - v_{x,t} r_t, \qquad \dot{r} = \frac{l_f F_{yf} - l_r F_{yr} - c_a r_t |r_t|}{I_z}.$$

Using forward Euler integration with step Δt,

$$x_{t+1} = x_t + \Delta t \, \dot{x}_t.$$

The measured lateral acceleration is

$$a_{y,t} = \dot{v}_{y,t} + v_{x,t} r_t + \eta_t,$$

where η_t is additive noise.

The adopted formulation relies on several simplifying assumptions. First, the vehicle is modeled using a planar bicycle representation, neglecting roll, pitch, and vertical dynamics. Second, the normal load at each axle is assumed constant and equal to its static distribution, which implies that load transfer effects due to acceleration, braking, and lateral dynamics are neglected. Third, the tire forces are assumed to depend only on slip angles and not on vertical load, camber, or combined slip effects. Finally, aerodynamic and suspension-induced variations of the vertical forces are not modeled explicitly.

These assumptions are clearly strong and would not be suitable for high-fidelity vehicle simulation. However, they are acceptable in the present work because the objective is not to obtain a practically deployable vehicle model, but to construct a controlled benchmark for comparing different modeling paradigms. The considered problem is therefore used as an analysis tool rather than a practical modeling target.

In particular, the mismatch is intentionally concentrated in the tire-force formulation, while vertical-load dynamics and load-dependent tire effects are neglected. This allows isolating the impact of the model structure and learning architecture without introducing additional factors.

3 Evaluated Models

This section describes the three model families evaluated in this work: a purely data-driven recurrent neural network, a physics-based bicycle model, and a hybrid physics-informed recurrent architecture. The goal is to assess how different levels of physical inductive bias affect lateral acceleration prediction.

3.1 Standard GRU Overview

A standard gated recurrent unit (GRU) is used as a purely data-driven baseline. Let $\mathbf{u}_k \in \mathbb{R}^d$ be the input at step k, $\mathbf{h}_k \in \mathbb{R}^n$ the hidden state, and $\mathbf{x}_k \in \mathbb{R}^m$ the output. The GRU dynamics are

$$\mathbf{z}_k = \sigma(W_z \mathbf{u}_k + U_z \mathbf{h}_{k-1} + \mathbf{b}_z)\,, \tag{1}$$

$$\mathbf{r}_k = \sigma(W_r \mathbf{u}_k + U_r \mathbf{h}_{k-1} + \mathbf{b}_r)\,, \tag{2}$$

$$\tilde{\mathbf{h}}_k = \tanh(W_h \mathbf{u}_k + U_h(\mathbf{r}_k \odot \mathbf{h}_{k-1}) + \mathbf{b}_h)\,, \tag{3}$$

$$\mathbf{h}_k = (\mathbf{1} - \mathbf{z}_k) \odot \mathbf{h}_{k-1} + \mathbf{z}_k \odot \tilde{\mathbf{h}}_k, \tag{4}$$

$$\mathbf{x}_k = W_{\text{out}} \mathbf{h}_k + \mathbf{b}_{\text{out}}. \tag{5}$$

All matrices and biases $\{W_z, U_z, \mathbf{b}_z, W_r, U_r, \mathbf{b}_r, W_h, U_h, \mathbf{b}_h, W_{\text{out}}, \mathbf{b}_{\text{out}}\}$ are learned from data.

3.2 Residual PhyGRU Overview

The PhyGRU model combines a physics-based state update with a learnable latent correction. Let $\mathbf{u}_k \in \mathbb{R}^d$ denote the input, $\boldsymbol{\xi}_k \in \mathbb{R}^{n_\xi}$ the physical state, $\boldsymbol{\ell}_k \in \mathbb{R}^{n_\ell}$ the latent state, and $\mathbf{h}_k = [\boldsymbol{\xi}_k^\top \ \boldsymbol{\ell}_k^\top]^\top \in \mathbb{R}^{n_\xi + n_\ell}$ the full recurrent state. The residual PhyGRU used in this work can be written as

$$\mathbf{z}_k = \sigma\left(W_z \begin{bmatrix} \mathbf{h}_{k-1} \\ \mathbf{u}_k \end{bmatrix} + \mathbf{b}_z\right), \tag{6}$$

$$\dot{\boldsymbol{\xi}}_{k-1}^{\text{phy}} = f_{\text{phy}}(\boldsymbol{\xi}_{k-1}, \mathbf{u}_k; \boldsymbol{\theta}_{\text{phy}}), \tag{7}$$

$$\boldsymbol{\xi}_k^{\text{phy}} = \boldsymbol{\xi}_{k-1} + \Delta t\, \dot{\boldsymbol{\xi}}_{k-1}^{\text{phy}}, \tag{8}$$

$$\dot{\boldsymbol{\ell}}_{k-1} = f_{\text{lat}}(\mathbf{h}_{k-1}, \mathbf{u}_k; \boldsymbol{\theta}_{\text{lat}}), \tag{9}$$

$$\boldsymbol{\ell}_k^{\text{cand}} = \boldsymbol{\ell}_{k-1} + \Delta t\, \dot{\boldsymbol{\ell}}_{k-1}, \tag{10}$$

$$\boldsymbol{\xi}_k^{\text{cand}} = \boldsymbol{\xi}_k^{\text{phy}} + \boldsymbol{\ell}_{k,\,1:n_\xi}^{\text{cand}}, \tag{11}$$

$$\tilde{\mathbf{h}}_k^{\text{res}} = \begin{bmatrix} \boldsymbol{\xi}_k^{\text{cand}} \\ \boldsymbol{\ell}_{k-1} \end{bmatrix}, \tag{12}$$

$$\mathbf{h}_k = (\mathbf{1} - \mathbf{z}_k) \odot \mathbf{h}_{k-1} + \mathbf{z}_k \odot \tilde{\mathbf{h}}_k^{\text{res}}, \tag{13}$$

$$\mathbf{x}_k = W_{\text{out}} \mathbf{h}_k + \mathbf{b}_{\text{out}}. \tag{14}$$

In this residual formulation, the latent branch acts as a local correction to the physical candidate state. The latent component is not used as an independent recurrent memory; instead, it modifies the physics-driven update through

$$\boldsymbol{\xi}_k^{\text{cand}} = \boldsymbol{\xi}_k^{\text{phy}} + \boldsymbol{\ell}_{k,\,1:n_\xi}^{\text{cand}}.$$

The learnable parameters are split into three groups:

$$\boldsymbol{\theta}_{\text{phy}} = \{m, I_z, c_f, c_r\}, \qquad \boldsymbol{\theta}_{\text{lat}}, \qquad \{W_z, \mathbf{b}_z, W_{\text{out}}, \mathbf{b}_{\text{out}}\}.$$

Here $\boldsymbol{\theta}_{\text{phy}}$ are the trainable coefficients of the bicycle prior, while $\boldsymbol{\theta}_{\text{lat}}$ and the gate/readout weights define the learned residual correction.

3.3 Linear Bicycle Prior

The physics prior is a bicycle model with linear tire forces. The state is

$$\mathbf{x}_k = \begin{bmatrix} v_{y,k} \\ r_k \end{bmatrix}, \qquad \mathbf{u}_k = \begin{bmatrix} \delta_k \\ v_{x,k} \end{bmatrix}.$$

The slip angles are

$$\alpha_{f,k} = \delta_k - \frac{v_{y,k} + l_f r_k}{v_{x,k}}, \qquad \alpha_{r,k} = -\frac{v_{y,k} - l_r r_k}{v_{x,k}}.$$

The linear tire law is

$$F_{yf,k} = c_f \alpha_{f,k}, \qquad F_{yr,k} = c_r \alpha_{r,k}.$$

The continuous-time bicycle dynamics are

$$\dot{v}_{y,k} = \frac{F_{yf,k} + F_{yr,k}}{m} - v_{x,k} r_k, \tag{15}$$

$$\dot{r}_k = \frac{l_f F_{yf,k} - l_r F_{yr,k}}{I_z}, \tag{16}$$

and the lateral acceleration output is computed as

$$a_{y,k} = \dot{v}_{y,k} + v_{x,k} r_k.$$

The learnable parameters in this prior are only

$$\boldsymbol{\theta}_{\text{phy}} = \{m, I_z, c_f, c_r\}.$$

The geometric quantities l_f and l_r are fixed, as is the integration step Δt. Therefore, the pure physics baseline estimates the vehicle response by learning just four scalar coefficients, whereas the PhyGRU learns those same physical coefficients jointly with the residual recurrent correction.

4 Methodology

For each training trajectory length in samples (maneuver duration $= dt\,T$; $dt = 0.05$)

$$T \in \{5, 25, 50, 100, 200, 250, 300, 500\}$$

the same experimental protocol is applied. For a fixed T, we generate 16 training sequences and 16 validation sequences from the nonlinear vehicle simulator described above. The input-output pairs are standardized using the training-set statistics, namely

$$\tilde{\mathbf{u}}_t = \frac{\mathbf{u}_t - \mu_u}{\sigma_u}, \qquad \tilde{y}_t = \frac{y_t - \mu_y}{\sigma_y}.$$

This normalization is applied consistently to the validation and test sets.

Three models are trained independently for each trajectory length: a fully data-driven GRU baseline with 13,121 learnable parameters, a physics-based linear bicycle model with only four learnable parameters $\boldsymbol{\theta}_{\text{phy}} = \{m, I_z, c_f, c_r\}$, and a residual PhyGRU hybrid model with 4,770 learnable parameters that combines the same bicycle prior with a learned latent correction and a gated recurrent update.

All models are optimized with mean squared error loss, using Adam. During training, validation loss is monitored at each epoch, and the checkpoint with minimum validation error

is retained. Early stopping is used with a patience of 50 epochs and maximum epochs fixed at 500. The selected model is then evaluated on 8 unseen test trajectories, each of length 1000, and the reported test score is the mean squared error on each trajectory.

A direct comparison between GRU and PhyGRU using the same number of learnable parameters is not necessarily fair. Although the parameter counts may be comparable, the PhyGRU architecture embeds additional inductive biases through the explicit physics prior and the discrete-time integration step. These components introduce structured dynamics that are not learned from data but are hard-coded into the model. Consequently, the effective modeling capacity of PhyGRU is larger than what is suggested by the number of learnable parameters alone. Therefore, even when the number of learnable parameters is matched, the PhyGRU benefits from a stronger structural prior and should be interpreted as a hybrid model with additional built-in knowledge rather than a purely parameter-matched alternative.

5 Results

The prediction task considered here is the estimation of lateral acceleration from steering input and longitudinal speed, which is a standard but nontrivial vehicle-dynamics problem. All three models are therefore required to learn a meaningful input-output mapping for the same physical quantity, namely the lateral response of the vehicle under dynamic excitation. This makes the comparison particularly relevant, because the task can be approached either by directly learning the temporal relationship from data, by using a mechanistic bicycle approximation, or by combining the two through a hybrid residual architecture, PhyGRU.

Table 1 summarizes the best validation error and the mean test MSE over the 8 test trajectories for every model and trajectory length. Figure 1 shows a clear dependence on the amount of temporal information available during training. For very short sequences ($T = 5, 25, 50$), the physics-only model generalizes best on average, despite its simplicity. For intermediate sequence lengths ($T = 100, 200, 250$), the residual PhyGRU becomes the most competitive model, benefiting from the combination of a mechanistic prior and a learned correction. For the longest trajectories ($T = 300, 500$), the pure GRU achieves the lowest mean test error, indicating that sufficient temporal context allows the fully data-driven model to outperform the hybrid alternatives in this setting.

An additional observation is that the best validation score does not always correspond to the best test performance, especially for the shortest trajectories. In particular, at $T = 5$ the PhyGRU attains the best validation loss, but its test error is substantially larger than that of the physics baseline, suggesting limited robustness in this low-data regime.

The parameter-efficiency of the physics model is evident, since it learns only four scalar coefficients, but this advantage is offset by model mismatch when the sequence length grows and the data become richer. The PhyGRU improves over the physics baseline for moderate and long trajectories by learning residual dynamics on top of the bicycle prior, while the GRU benefits most from the largest training horizons and eventually achieves the best overall generalization.

Larger latent-state configurations for the PhyGRU were not investigated. Although increasing the latent dimension would improve model expressiveness, it would also reduce the influence of the physics prior. For sufficiently large latent spaces, the hybrid architecture would likely collapse into a fully data-driven model, with the residual branch dominating the dynamics. Since the goal of this work is to assess hybrid models that retain a meaningful physical inductucive bias, such configurations were considered less informative and therefore not explored.

Trajectory T [samples]	Model	Test 1	Test 2	Test 3	Test 4	Test 5	Test 6	Test 7	Test 8
5	Physics	0.131879	0.013462	0.054859	0.183704	0.087344	0.092531	0.012027	0.053155
	PhyGRU	2.403949	1.561770	1.237667	2.496377	2.903115	2.142429	1.522446	1.347111
	GRU	1.003230	1.378271	1.781120	1.913496	1.857367	1.035210	1.435201	1.821006
25	Physics	0.143792	0.015089	0.061132	0.206528	0.098609	0.102748	0.013595	0.059734
	PhyGRU	0.172212	0.226971	0.287752	0.412508	0.388468	0.169702	0.242350	0.300726
	GRU	0.115613	0.185949	0.277651	0.340168	0.329589	0.117391	0.193036	0.286061
50	Physics	0.197151	0.020518	0.084408	0.279954	0.134357	0.142719	0.018329	0.082172
	PhyGRU	0.227401	0.277954	0.289345	0.408805	0.383921	0.216120	0.304518	0.301081
	GRU	0.235150	0.332997	0.417286	0.595088	0.508088	0.231319	0.361917	0.445223
100	Physics	0.174279	0.017561	0.076595	0.243298	0.117152	0.125075	0.017081	0.069904
	PhyGRU	0.049943	0.057862	0.038020	0.101268	0.042286	0.042194	0.059110	0.032349
	GRU	0.099763	0.165764	0.205146	0.298552	0.222990	0.094593	0.172203	0.224003
200	Physics	0.050619	0.005423	0.021011	0.072097	0.033755	0.036995	0.004736	0.020738
	PhyGRU	0.015504	0.007580	0.010976	0.022551	0.011946	0.010605	0.007425	0.011388
	GRU	0.024739	0.028831	0.040090	0.026054	0.018563	0.024370	0.029453	0.038419
250	Physics	0.056561	0.005329	0.023611	0.078815	0.038764	0.039815	0.005765	0.022972
	PhyGRU	0.032125	0.010177	0.015776	0.016824	0.020032	0.021644	0.010647	0.016349
	GRU	0.036294	0.035293	0.042321	0.028585	0.026709	0.031382	0.034974	0.042645
300	Physics	0.063735	0.006562	0.027364	0.090358	0.042424	0.045830	0.005881	0.025622
	PhyGRU	0.026067	0.007654	0.009687	0.022151	0.027725	0.016155	0.007611	0.010581
	GRU	0.013376	0.004567	0.006621	0.008329	0.013518	0.008420	0.004751	0.007014
500	Physics	0.097626	0.009952	0.041597	0.137938	0.065048	0.070028	0.009867	0.039632
	PhyGRU	0.039585	0.006440	0.014701	0.049927	0.026714	0.021685	0.006751	0.015210
	GRU	0.018076	0.004112	0.007761	0.024047	0.016967	0.010924	0.003967	0.007579

Table 1: Test MSE for each trajectory length.

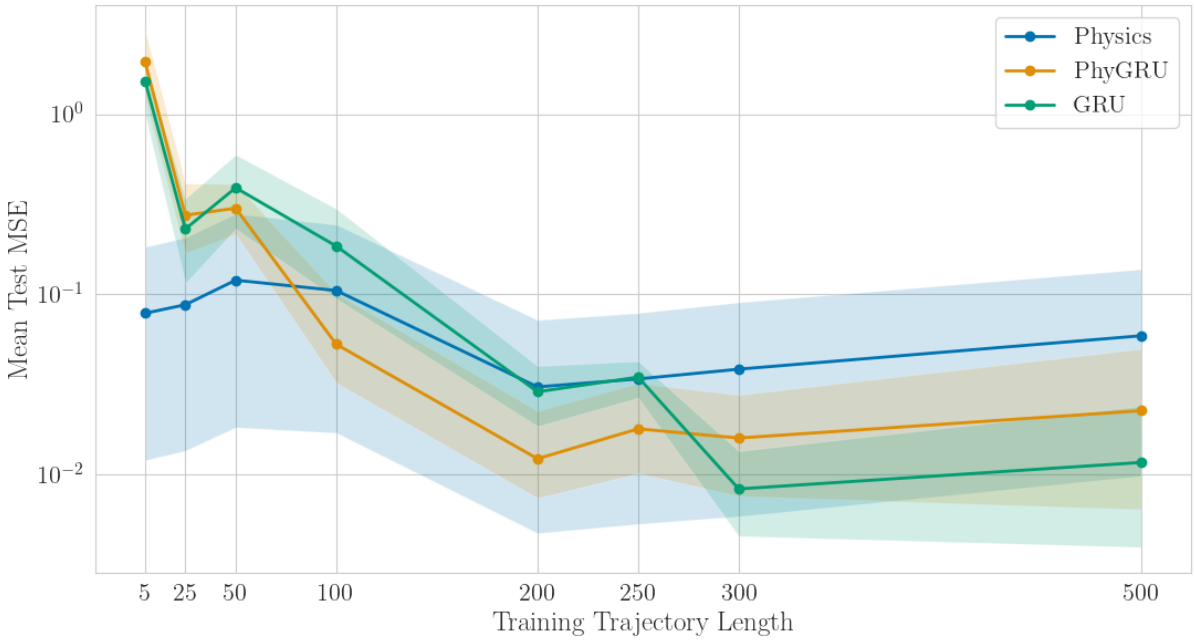

Figure 1: Mean test MSE over different training set size, with min-max bands.